

HEL, Poland

WMO: 121350

Lat: 54.6036N

Lon: 18.8119E

Elev: 9

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	19.3	10.1	9.2	16.6	13.9	11.2	21.0	29.2	40.7	25.1	39.1	8.8	180	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	78.1	68.7	75.3	67.0	72.8	65.7	70.2	76.0	68.4	73.5	66.7	71.3	7.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	103.3	73.9	66.5	97.4	71.5	64.8	92.0	69.7	34.2	76.0	32.7	73.8	31.4	71.4	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	18.5	16.4	DB	12.6	83.4	5.4	3.8	8.7	86.1	5.5	88.4	2.4	90.5	-1.5	93.2	(4)
(5)			WB	11.5	73.3	5.2	2.1	7.8	74.8	4.8	76.0	1.8	77.1	-1.9	78.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	32.2	32.9	36.8	44.2	52.6	59.8	64.7	65.0	58.4	49.5	41.7	35.7
	DBStd	12.95	6.86	6.42	5.39	5.82	6.06	4.82	4.34	4.04	4.38	5.22	5.31	6.50
	HDD50	2439	551	478	411	191	38	0	0	0	2	74	251	443
	HDD65	6364	1016	898	875	623	386	169	59	49	202	480	699	908
	CDD50	1669	0	0	0	18	120	295	457	466	253	59	1	0
	CDD65	118	0	0	0	0	2	13	50	50	3	0	0	0
	CDH74	383	0	0	0	1	10	60	184	123	5	0	0	0
	CDH80	42	0	0	0	0	0	8	25	9	0	0	0	0
Wind	WSAvg	8.8	10.2	9.7	9.0	8.0	7.6	7.9	7.7	7.7	8.3	9.4	9.9	10.4
Precipitation	PrecAvg	22.8	1.3	1.1	1.2	1.2	2.0	2.2	3.1	2.6	2.4	2.1	1.9	1.6
	PrecMax	28.1	3.0	2.0	2.2	2.8	3.6	5.0	6.6	5.9	5.7	4.8	4.1	2.7
	PrecMin	16.4	0.2	0.1	0.2	0.1	0.7	0.2	0.4	0.2	0.6	0.5	0.4	0.7
	PrecStd	3.0	0.7	0.5	0.5	0.7	0.9	1.1	1.4	1.5	1.3	1.1	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	46.6	53.1	67.5	75.2	80.5	83.3	80.9	73.9	63.0	54.1	49.7
		MCWB	44.5	43.9	46.6	56.5	65.0	68.8	71.5	70.5	65.4	59.0	51.9	47.4
	2%	DB	44.4	44.4	48.8	61.3	70.3	75.0	78.6	77.2	69.4	60.4	51.6	47.4
		MCWB	42.3	42.1	43.9	53.1	61.8	65.7	69.3	68.5	63.4	56.8	49.7	45.4
	5%	DB	42.5	42.7	46.0	57.0	66.3	71.3	75.5	74.6	67.0	58.6	50.0	45.5
		MCWB	40.7	40.6	42.4	50.4	58.8	63.4	67.6	67.0	61.5	55.7	48.1	43.9
	10%	DB	40.7	41.1	43.9	53.4	62.7	68.3	72.7	72.3	64.9	56.8	48.5	43.8
		MCWB	39.1	39.4	41.2	48.2	56.8	61.7	66.2	65.9	60.3	54.3	46.7	42.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	44.7	48.0	58.3	65.7	70.2	73.3	72.4	67.0	59.9	52.6	47.8
		MCDB	47.1	46.0	51.4	66.0	73.1	78.3	80.1	78.5	71.8	62.2	53.4	49.0
	2%	WB	42.7	42.6	44.8	53.9	62.6	66.9	70.9	69.8	64.3	57.7	50.1	46.0
		MCDB	44.1	44.0	47.6	60.2	68.6	73.1	77.0	75.2	68.0	59.7	51.3	47.1
	5%	WB	40.9	40.9	43.0	51.1	59.9	64.6	69.0	68.2	62.6	56.2	48.4	44.0
		MCDB	42.3	42.3	45.5	56.3	65.1	69.9	73.9	73.1	66.0	58.1	49.9	45.2
	10%	WB	39.2	39.4	41.3	48.7	57.2	62.5	67.1	66.7	61.0	54.6	47.0	42.3
		MCDB	40.5	41.0	43.6	52.7	62.0	67.2	71.5	70.9	64.1	56.6	48.3	43.7
Mean Daily Temperature Range	5% DB	MDBR	5.1	6.0	8.8	13.2	14.3	13.2	12.2	11.9	9.4	7.3	5.4	5.1
		MCDBR	5.7	7.0	12.4	19.2	19.1	17.9	16.7	15.1	12.1	8.9	6.2	5.8
		MCWBR	5.3	5.9	8.5	11.9	10.9	10.2	8.9	8.1	7.5	6.8	5.7	5.6
	5% WB	MCDBR	5.7	7.0	11.5	17.8	18.2	16.8	15.3	14.0	11.6	8.7	6.1	5.8
MCWBR		5.5	6.2	8.4	11.6	11.0	10.4	8.7	8.3	7.6	6.9	5.8	5.7	
Clear-Sky Solar Irradiance	taub	0.313	0.341	0.366	0.409	0.399	0.396	0.427	0.419	0.379	0.361	0.346	0.312	
	taud	2.298	2.305	2.298	2.231	2.333	2.376	2.306	2.329	2.418	2.437	2.329	2.294	
	Ebn at Noon	197	233	254	258	268	270	258	253	249	225	182	170	
	Edh at Noon	18	25	32	39	37	36	38	35	28	23	18	15	
All-Sky Solar Radiation	RadAvg	145	356	757	1338	1744	1830	1640	1399	986	495	187	105	
	RadStd	19	55	85	161	168	149	180	136	107	77	24	14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-223	+133	N/A
(47) Regional (22 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-243	+119	N/A

Nomenclature: See separate page